

A Literature-Implied Answer to the Vector-Valued Lipschitz Octahedrality Question

arXiv automatic math run `fa_banach_001`

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Status

Status: literature-implied answer. This packet records that a question from Langemets–Rueda Zoca, arXiv:1905.09061, is answered by combining later published/preprint results. It is not claimed as an original proof.

Source Question

The source paper is J. Langemets and A. Rueda Zoca, *Octahedral norms in duals and biduals of Lipschitz-free spaces*, arXiv:1905.09061. Problem 1.3 in that paper asks:

Let M be a metric space such that $\text{Lip}_0(M)$ is octahedral and let X be a non-trivial Banach space. Is then the space $\text{Lip}_0(M, X)$ octahedral?

Answer

Yes. The answer is affirmative.

Identification

The first ingredient is A. Rueda Zoca, *Banach spaces which always produce octahedral spaces of operators*, arXiv:2207.08717. Example 4.3 states, using arXiv:1905.09061, Proposition 3.3, that if $\text{Lip}_0(M)$ is octahedral, then $\text{Lip}_0(M)$ is c_0 -octahedral and consequently universally octahedral. In particular,

$$L(Y, \text{Lip}_0(M))$$

is octahedral for every Banach space Y .

The second ingredient is R. Medina and A. Rueda Zoca, *A characterisation of the Daugavet property in spaces of vector-valued Lipschitz functions*, arXiv:2305.05956. Theorem 4.2 characterizes universal octahedral domains: for a Banach space Z , the following are equivalent:

$$L(Z, Y) \text{ is octahedral for every Banach space } Y$$

and

$$Z^* \text{ is universally octahedral.}$$

The theorem is stated in a slightly stronger form, allowing all subspaces of $L(Z, Y)$ that contain the finite-rank operators.

Apply Theorem 4.2 with $Z = \mathcal{F}(M)$. Since $\mathcal{F}(M)^* = \text{Lip}_0(M)$ and $\text{Lip}_0(M)$ is universally octahedral by the preceding paragraph, $L(\mathcal{F}(M), X)$ is octahedral for every Banach space X . Finally, the standard linearization isometry gives

$$\text{Lip}_0(M, X) \cong L(\mathcal{F}(M), X).$$

Therefore $\text{Lip}_0(M, X)$ is octahedral for every non-trivial Banach space X , answering Problem 1.3 affirmatively.

Scope and Limitations

This packet only records the implication chain above. It does not settle the broader tensor-product question asking whether $X \widehat{\otimes}_\pi Y$ is octahedral whenever X is octahedral.

References

- J. Langemets and A. Rueda Zoca, *Octahedral norms in duals and biduals of Lipschitz-free spaces*, arXiv:1905.09061.
- A. Rueda Zoca, *Banach spaces which always produce octahedral spaces of operators*, arXiv:2207.08717.
- R. Medina and A. Rueda Zoca, *A characterisation of the Daugavet property in spaces of vector-valued Lipschitz functions*, arXiv:2305.05956.